

Spatial Fidelity Maps: A Query-Time, Ground-Truth-Free Signal for Approximate Nearest-Neighbor Retrieval

Abstract

Approximate nearest-neighbor (ANN) systems trade retrieval quality for latency, storage, and energy, but the standard way of quantifying that trade — *candidate fidelity*, the overlap between an approximate index’s results and the exact top- k — can only be measured by running the exact-KNN search the approximate index exists to avoid. This is impractical at query time and self-defeating as a deployment-time signal. We propose the *spatial fidelity map*: a small, offline-built lookup structure that predicts candidate fidelity from a query’s location in the corpus alone, requiring only a query’s nearest landmark rather than a full retrieval or an exact reference. We state the two empirical premises the map depends on (that fidelity varies smoothly across the embedding space, and that it must be estimated separately per index configuration), and we specify a complete, self-contained verification methodology — five targeted experiments (H1–H5) and a concrete offline construction pipeline — for testing whether these premises hold before the map is trusted in a live system. This is a study-design paper: we specify what must be built and measured, not yet report results.

1 Introduction

Modern retrieval systems deploy approximate nearest-neighbor (ANN) indexes — HNSW [8], IVFPQ [2, 4], OPQ-rotated variants [3], DiskANN [6] — because exact search over corpora of millions of vectors is too slow and too expensive to run at query time. The price of that approximation is usually reported as *candidate fidelity*: the fraction of a query’s true top- k nearest neighbors that the approximate index actually returns. Fidelity is the standard way of answering “how much did we give up by not running exact search,” and it directly predicts downstream retrieval quality after any reranking or fusion stage is applied.

The problem this paper addresses is that fidelity, as conventionally computed, requires running an exact-KNN pass over the corpus for the query in question — precisely the computation the approximate index was built to avoid. Every existing use of fidelity as an evaluation metric, including in large-scale ANN benchmarking, computes it offline, against a materialized exact reference, and never at deployment time. If a retrieval system wants to reason about fidelity *while serving a query* — for example, to decide whether a cheap index configuration is trustworthy enough for this particular query,

or whether to fall back to a more expensive one — no existing mechanism supplies that signal without contradicting the reason ANN indexing was adopted in the first place.

Idea. We propose that a query’s fidelity under a given index configuration is, to a useful first approximation, a function of *where the query sits in the corpus*, not of the full retrieval that configuration would perform. A query in a dense region of the embedding space is typically well approximated even under aggressive quantization; a query in a sparse region is not, largely independent of the query’s specific semantic content. If this holds, then locating a query — finding only its single nearest landmark in a small, precomputed set — can substitute for running the configuration at all when predicting how much to trust it.

Goal of this study. It is worth stating precisely what accurate prediction is *for* here, since it shapes what “accurate enough” means. In deployment, a corpus is indexed under several methods and configurations, and when a query arrives the system must decide which configuration to use. That decision is not “which configuration has the highest CF” — taken literally, the answer is almost always the most expensive configuration (or exact search), and no predictor is needed to know that. The decision that actually matters is cost-constrained: among the configurations predicted to be *good enough*, use the cheapest. In the larger research program this study is part of, that decision takes the specific form of a *fidelity floor* τ : run the cheapest configuration predicted to achieve $\text{CF}@k \geq \tau$, escalating only if it appears to fall short. The full cost model — how latency, storage, and energy combine into a single budget — is deliberately left to that parent work and is not defined here; the fidelity floor is the one part of the cost/quality policy that is already pinned down, and it is enough to give this study a concrete target. The map’s job, then, is not to rank configurations by absolute CF but to answer, per query and without an exact-KNN reference, the threshold question “does configuration A_j clear τ ?” accurately enough that the cheapest clearing configuration is chosen correctly. This has a direct consequence for evaluation: prediction accuracy matters most *near* τ , where an estimation error can flip the decision, and matters little far from it, where a configuration is classified correctly even by a coarse estimate. We return to this when defining metrics (Section 5).

Contributions. This is a study-design paper. Concretely:

1. We formalize candidate fidelity and the problem of estimating it without an exact-KNN reference (Section 3).
2. We propose the spatial fidelity map: its construction, its query-time lookup mechanism, and the two empirical premises — *smoothness* and *index-specific degradation* — it depends on (Section 4).
3. We specify a complete verification methodology of five targeted experiments, H1–H5, each isolating one part of the map’s mechanism or one of its premises, so that the hypothesis is tested directly rather than only inferred from downstream accuracy (Section 5).
4. We give a concrete, storage-level offline pipeline for building the map at corpus scale (Section 7), so the verification methodology above is immediately actionable rather than aspirational.

2 Background and Related Work

Graph-based indexes such as HNSW [8] and DiskANN [6], and quantization-based indexes such as IVFPQ [2, 4] and its OPQ-rotated variant [3], are the dominant approaches to approximate similarity search at scale, surveyed for example in [9] and implemented at scale in libraries such as FAISS [5]. In all of this work, fidelity or recall is reported *as an evaluation metric*: computed once, offline, against a materialized exact-KNN reference, and used to compare index configurations against each other. None of this literature proposes estimating fidelity *without* that reference, which is the problem this paper addresses.

The premise that a query’s local neighborhood density predicts search difficulty is not new in spirit: local intrinsic dimensionality and hubness are well-studied phenomena describing how approximate-search difficulty varies across a corpus. What has not, to our knowledge, been proposed or tested is repurposing this kind of local-geometry signal as an explicit, queryable, deployment-time predictor of fidelity for a specific index configuration — which is precisely what the spatial fidelity map operationalizes below.

This paper reuses large-scale exact-KNN and fidelity measurement infrastructure we have developed in prior work evaluating cross-modal retrieval pipelines at LAION scale [1], described concretely in Section 6.

3 Problem Definition

For a query q , corpus C , and ANN configuration A (an index family together with its parameters, built over an embedding of C), define candidate fidelity at depth k as

$$\text{CF@}k(q, A) = \frac{|\text{Top-}k_{\text{KNN}}(q) \cap \text{Top-}k_A(q)|}{k},$$

the overlap between the exact top- k neighbors of q and the top- k that A actually returns. Computing $\text{CF@}k$ as defined requires $\text{Top-}k_{\text{KNN}}(q)$: an exact, corpus-wide nearest-neighbor pass, $O(|C|)$ per query. This is exactly the computation an approximate index exists to avoid, so $\text{CF@}k$ cannot be computed at deployment time without contradicting the point of using A in the first place.

We instead pose the *estimation* problem: given q and A , and a set of signals $\phi(q, A)$ computable *without* access to $\text{Top-}k_{\text{KNN}}(q)$, construct $\hat{f}$ such that $\hat{f}(\phi(q, A)) \approx \text{CF@}k(q, A)$. This paper studies one specific, minimal instantiation of ϕ : the query’s location relative to a small, pre-computed set of landmarks, obtained from retrieving only q ’s single nearest neighbor among those landmarks rather than any part of A ’s own top- k .

4 The Spatial Fidelity Map

4.1 Core Hypothesis

H0 (working hypothesis). *A query’s candidate fidelity under configuration A is predictable, to a useful approximation, from the local density of the corpus around the query’s location alone — without running A , and without computing the query’s own exact top- k .*

This is stated as a hypothesis, not an assumption: Section 5 exists specifically to test it before it is relied upon.

4.2 Construction

Given a corpus C embedded under a fixed model, and a catalog of index configurations $\{A_1, \dots, A_m\}$ built over that embedding:

1. **Tile the space.** Choose a set of landmarks $L = \{\ell_1, \dots, \ell_K\}$, $K \ll |C|$, by a single dedicated k -means pass over C . This L is the map’s own tiling of the embedding space: it is built once per model and *shared across every index configuration in that model’s catalog*, so that the (ℓ, A_j) keying below compares the same physical location across index families. We deliberately *do not* reuse any IVF-family index’s internal coarse-quantizer centroids as L : those differ from one configuration to the next (different `nlist`, and OPQ-rotated variants quantize in a rotated space), so borrowing them would make L configuration-dependent and the keying ablation (H3) ill-posed. A dedicated pass is also the natural choice given that H4 rebuilds L at several values of K regardless.
2. **Sample and label.** Draw a query sample Q , compute $\text{Top-}k_{\text{KNN}}(q)$ once per $q \in Q$ (the one unavoidable exact-KNN pass, paid offline and only once, reused by every A_j), and compute $\text{CF@}k(q, A_j)$ for every (q, A_j) pair by running each configuration against Q .

3. **Assign.** For each $q \in Q$, find its nearest landmark $\ell(q) = \arg \min_{\ell} \|q - \ell\|$, and record the distance to the second-nearest landmark as well (used by H2 below).
4. **Aggregate.** For every (ℓ, A_j) pair, aggregate $\text{CF@}k(q, A_j)$ over all q with $\ell(q) = \ell$: store the mean, standard deviation, and sample count.

4.3 Query-Time Lookup

At serving time, for a new query q^* and a candidate configuration A_j : find $\ell(q^*)$ via an $O(K)$ scan over L (thousands of points, not millions), and read off the precomputed $(\ell(q^*), A_j)$ cell. Because L is a single shared tiling rather than any index’s internal quantizer, this scan is one small, fixed, index-independent cost per query — paid once and shared across every configuration under consideration, not repeated per configuration — and it remains available *before* any A_j has actually been queried in full.

4.4 Two Premises to Verify

P1 (Smoothness). The map is only useful if fidelity varies slowly enough across the space that $\ell(q)$ ’s aggregated statistics are representative of q ’s own neighborhood, not merely of the landmark’s. This is plausible away from cluster boundaries; a query sitting at the edge of a dense region can have a nearest landmark whose own neighborhood is dense while the query’s true top- k reaches into sparser territory the landmark does not represent.

P2 (Index-specific degradation). Local density is a property of the embedding geometry alone, independent of which index family is eventually queried — but how much a given density translates into fidelity *loss* is index-family specific (IVFPQ’s degradation is dominated by quantization distortion; HNSW’s by graph connectivity and hub effects). The map must therefore be keyed by (ℓ, A_j) pairs, not by ℓ alone, or it risks conflating degradation mechanisms that do not actually agree with each other.

5 Verification Methodology

We verify H0 directly, via P1 and P2, rather than only inferring it from the map’s aggregate downstream accuracy. We first define the comparison points, then the five experiments.

5.1 Comparison Points

- **Global-Average (GA).** Predict the same value for every query under a given A_j : the mean of $\text{CF@}k(q, A_j)$ over all $q \in Q$, ignoring location entirely. This is the trivial baseline any location-aware signal must beat.
- **Density-Only (DO).** A landmark-keyed but *configuration-independent* variant: aggregate a corpus-geometry density score at each landmark (e.g., mean

distance to its k nearest corpus points) with no label information, combined with GA’s per-configuration average via a simple fitted offset. This isolates whether *configuration-specific* labeling (our full map) is doing real work beyond generic density.

- **Spatial Map (SM), proposed.** The (ℓ, A_j) -keyed map from Section 4.2.
- **Oracle.** The true $\text{CF@}k(q, A_j)$ itself — not deployable (requires the exact-KNN pass), used only to generate labels and to bound achievable accuracy.

5.2 H1: Error-versus-Distance Diagnostic

Plot SM’s prediction error against q ’s distance to $\ell(q)$, and separately against the gap between distances to the first- and second-nearest landmarks (a proxy for boundary proximity). P1 predicts error should grow with both quantities; a flat relationship would indicate the map is not capturing meaningful spatial structure at all, regardless of its aggregate accuracy.

5.3 H2: Boundary Stress Test

Partition held-out queries into *interior* (nearest landmark much closer than second-nearest) and *boundary* (comparably close) subsets, and report SM accuracy separately for each. P1’s failure mode predicts a visible accuracy gap between the two subsets; this directly tests the boundary-degradation concern rather than letting it average out and disappear inside an aggregate metric.

5.4 H3: Keyed vs. Shared-Across-Configurations Ablation

Compare SM ((ℓ, A_j) -keyed) against a variant keyed by landmark alone and shared across index families, on at least two structurally different families (e.g., HNSW and IVFPQ) built over the same embedding. P2 predicts the shared variant should perform visibly worse, since it conflates two different degradation mechanisms at the same location.

5.5 H4: Landmark Resolution Sweep

Vary K (the number of landmarks) from coarse to fine, and report the resulting accuracy, storage, and lookup-cost trade-off. Accuracy should improve with finer tiling up to the point where individual landmarks no longer have enough nearby sampled queries to estimate fidelity reliably, after which per-cell variance should increase and accuracy should degrade or plateau.

5.6 H5: Does Location Help at All?

Compare SM directly against GA and DO on the same held-out query set. This is the most fundamental check: if SM

does not clearly outperform GA, the entire premise that location predicts fidelity better than a single global number is unsupported, independent of any of the more detailed diagnoses in H1–H4.

5.7 Metrics

For each comparison, report Spearman correlation and mean absolute error between predicted and true $CF@k$, together with calibration curves (predicted vs. observed fidelity, binned) — correlation alone can look favorable while the estimator is still poorly calibrated, and a downstream fidelity-floor policy would need the latter, not only the former.

These aggregate metrics measure the map as a *CF regressor*, but the goal of Section 1 is a *decision*: does each configuration clear a fidelity floor τ , and is the cheapest clearing configuration chosen correctly? We therefore additionally report decision-oriented metrics evaluated directly against that threshold. Because the full cost model, and hence the exact floor, is deferred to the parent work, we sweep τ across a plausible range (e.g., 0.7, 0.8, 0.9) rather than committing to one value, and report at each: (i) *threshold-decision accuracy* — how often the map correctly classifies a (q, A_j) pair as above or below τ (equivalently, precision/recall of the “clears τ ” decision), which unlike MAE charges error only where it can change the decision; and (ii) *realized CF regret* — when the map’s floor-based selection differs from the correct one, the true $CF@k$ of the configuration the map would choose minus that of the configuration the oracle would choose, so a wrong pick between two near-tied configurations is penalized far less than a genuine miss. Sweeping τ shows whether the map remains decision-useful across whatever floor the eventual cost model settles on, rather than tuning the evaluation to one assumed policy; and reporting regret rather than only decision accuracy reflects that not all wrong decisions cost the same amount of fidelity.

6 Experimental Setup

6.1 Corpus and Models

We build the map on a 7-million-image subset of LAION [10], following the same scale used in our prior evaluation infrastructure [1], together with benchmark-scale query sets from MS-COCO [7] and Flickr30K [11] for smaller, faster verification passes during development. Because LAION is a corpus of image–text pairs embedded into a joint space, the query and index modalities need not match, and the modality gap between them (Section 6.3) means the map must be verified under each modality configuration rather than only in aggregate. We use four embedding models spanning three architecture families, each with its own embedding space, landmark set, and index catalog — models are never compared to each other by the map, only used to test whether H1–H5’s conclusions replicate

across independently built spaces. Two contrastive single-vector dual-encoders (SigLIP 2, and an OpenCLIP model trained on LAION itself, i.e. native to the corpus) and one LLM-backed unified single-vector retriever (Qwen3-VL-Embedding) form the primary set; a fourth, gap-reducing unified model (Jina Embeddings v4) adds a second point on the modality-gap axis. All four are single-vector, keeping the landmark tiling and the “query’s location” well defined. A late-interaction / multi-vector retriever (e.g. ColPali), for which a query is a bag of token vectors and “location” is no longer a single point, is the natural but hardest case; we treat it as a gated final step, run only once H1–H5 support the premises on the single-vector models, and out of scope for the initial pilot. We stage the evaluation as a 1M-pair pilot drawn as a stratified subsample of the 7M pool (Section 6.4), treated as a go/no-go gate — if H5 fails there, H0 has already failed in the cheaper setting — before committing to the full 7M run.

6.2 Index Catalog

For each model, we build all three families — HNSW, IVFPQ, and O-IVFPQ [8, 2, 3]. Within each family we build several configurations chosen to *span* the candidate-fidelity axis (a low-, moderate-, and high-fidelity operating point), rather than a single per-family default: because the map’s job is to predict $CF@k$, it can only be verified where CF actually varies, so configurations that bracket low through high fidelity are the ones that make H1–H5 discriminating. Concrete operating points are taken from the parameter sweeps of our prior large-scale study [1], which measured $CF@100$ for exactly these families on this corpus. The identical configuration set is used for every model, so the H3 keying ablation is comparable across models; the H4 sweep additionally varies the landmark count K (not the index parameters), holding the configuration set fixed.

6.3 Modality Configurations

LAION is a corpus of image–text pairs, and the embedding models we use map both modalities into a single joint space per model. This raises a question the map’s construction (Section 4.2) leaves implicit: which modality populates the index, and which modality the queries are drawn from. Because image and text embeddings occupy systematically offset regions of a joint CLIP-style space (the modality gap), and because LAION pairs are themselves pre-filtered by cross-modal similarity, the query and index modalities cannot be assumed interchangeable, and H0 must be tested under each combination separately rather than only in aggregate. We therefore evaluate the map under all four query/index modality configurations: text-query/text-index, image-query/image-index, text-query/image-index, and image-query/text-index. The two same-modality configurations isolate the geometry of each embedding distribution on its own; the two cross-modality configurations expose the

map to the modality gap and to any asymmetry between the two cross directions.

Our verification target is pairing-blind: $CF@k(q, A_j)$ is a property of a query relative to the index geometry (Section 3) and does not reference the query’s paired partner. The configurations differ only in which modality plays the index role and which plays the query role; the H1–H5 experiments (Section 5) are run, and reported, once per configuration.

6.4 Data Splits and Sampling

Every configuration uses three disjoint roles, mirroring the three stages of the map’s life cycle: an *index set* that populates the searchable corpus and against which $CF@k$ is defined; a *map-construction set* of sample queries Q used to estimate the per-landmark statistics of Section 4.2; and a *test set* of held-out queries on which every H-experiment evaluates the map’s predictions. All three must be disjoint: the map-construction and test sets must not overlap, or the map is evaluated on the very points that calibrated it; and both query sets must be disjoint from the index set, since a query that is also an index point retrieves itself at distance zero and distorts both the calibration statistics and the test.

A single pair-level split, shared across configurations. Rather than split each modality independently per configuration, we partition the underlying LAION image–text *pairs* once, into three disjoint parts $\mathcal{A}, \mathcal{B}, \mathcal{C}$, and reuse this one partition across all four configurations. $\mathcal{A}$ always supplies the index (its text side for a text index, its image side for an image index); $\mathcal{B}$ always supplies the map-construction queries; $\mathcal{C}$ always supplies the test queries; in each case the modality is read out as the configuration requires. The index is thus always drawn from $\mathcal{A}$ alone — never the full corpus — so $CF@k$ is defined relative to the same reference geometry ($\mathcal{A}$) in every configuration.

Splitting at the pair level is stricter than the pairing-blind $CF@k$ strictly requires, and we adopt it deliberately as a control. Because $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are disjoint as pairs, the paired partner of any test query in $\mathcal{C}$ also lies in $\mathcal{C}$, never in the index $\mathcal{A}$. Consequently *no query’s paired object is ever present in the index, in any of the four configurations*. This forecloses the objection that the cross-modality configurations are advantaged relative to the same-modality ones — for instance, that a text query already has a strongly matching image partner sitting in the image index. Under our split no such partner is indexed, and this holds identically across the 2×2 , so differences in map accuracy trace to modality rather than to an asymmetric partner-in-index confound.

Stratified sampling. A purely uniform draw would populate $\mathcal{A}, \mathcal{B}, \mathcal{C}$ in proportion to the corpus’s own density, oversampling dense regions and leaving sparse regions — precisely where the smoothness premise P1 is most likely to fail — underrepresented in both the map-construction and test sets. Since P1 and its boundary failure mode (H2) are exactly what we intend to stress, we instead sample in a density-aware way.

Density, however, is not a single well-defined quantity here: the image and text embeddings of a model occupy different regions of its joint space and have different dense and sparse regions, so a pair may lie in a dense part of image space and a sparse part of text space. Because our split is shared across all four configurations, and the smoothness premise must hold in whichever modality plays the *query* role — text space in the text-query configurations, image space in the image-query ones — stratifying on either modality alone would guarantee balanced coverage only for the two configurations that query that modality, and leave the other two stress-tested only incidentally. We therefore stratify *jointly* on both modalities. We run one k -means pass in image space and one in text space, label each pair by the pair of clusters it falls into, and form strata from these joint labels; we then draw $\mathcal{A}, \mathcal{B}, \mathcal{C}$ from within each joint stratum. A pair that is boundary-like or sparse in *either* modality is thus guaranteed representation in all three sets, so the region where P1 is most likely to fail is covered for every configuration rather than only for the two that happen to query the stratified space.

Stratification is performed once, at the pair level, before any modality read-out, so that the same pairs occupy the same role in every configuration. It is important to note that this stratification k -means is a *distinct, and much coarser*, clustering than the landmark k -means of Section 4.2: the strata exist only to balance the draw across dense and sparse regions, so they need coarse, heavily-populated buckets, whereas the landmarks L estimate per-cell CF and need fine cells ($K \approx \sqrt{|\mathcal{A}|} \approx 1,000$ at 1M). Concretely, we use a per-modality cluster count of $k \approx 25$ (range 20–50) in each of image space and text space, giving a joint cross-product of at most ~ 400 – $2,500$ strata — of which many will be near-empty (a cluster dense in one modality paired with one sparse in the other), so the effective count is smaller and only non-empty strata are drawn from. At the 1M pilot this leaves on the order of hundreds of pairs per populated stratum, ample to draw the 85/10/5 split of $\mathcal{A}, \mathcal{B}, \mathcal{C}$ within each. We verify this by histogramming pairs-per-joint-stratum and coarsening k if the sparse tail contains strata too small to split; the stratification k is a sampling device only and, unlike the landmark K , is not swept and never appears in H1–H5. The number of pairs assigned to each role is held identical across configurations, so sample size is never a confound between them.

Split sizes. The binding constraint on sizing is the number of map-construction queries that fall in each landmark cell: the served map stores a per-cell mean of $CF@k$ over the $\mathcal{B}$ queries assigned to that landmark, so a cell with too few queries yields a noisy, unreliable estimate. We size $\mathcal{B}$ so that this per-cell count clears a floor of roughly 30–50 queries in the *sparse* cells — not merely on average — since the stratified draw deliberately populates sparse and boundary regions, which are exactly the cells P1 and H2 stress and therefore the ones whose estimates must not be starved. The index set $\mathcal{A}$, by contrast, is sized to be a realistic corpus

rather than to any per-cell floor: it defines the reference geometry against which $\text{CF}@k$ is computed, and takes the bulk of the pairs. The test set $\mathcal{C}$ need only be large enough for stable held-out error and for the interior/boundary partition of H2 to carry real counts on each side.

Concretely, for an initial 1M-pair pilot with $K \approx 1,000$ landmarks ($K \approx \sqrt{|\mathcal{A}|}$, the center of the H4 sweep), we allocate $|\mathcal{A}| \approx 850,000$, $|\mathcal{B}| \approx 100,000$, and $|\mathcal{C}| \approx 50,000$ (roughly 85/10/5), giving on the order of 100 construction queries per cell on average and leaving sparse cells above the floor. Before trusting the map we histogram the realized $\mathcal{B}$ -per-cell counts rather than only their mean; cells in the bottom decile falling below the floor are treated as an H4 finding about landmark resolution, not silently averaged away. Scaling to the full 7M-pair corpus, the three roles do *not* scale proportionally: $\mathcal{A}$ scales with the corpus ($\gtrsim 6\text{M}$), but K grows only sublinearly ($\sqrt{7\text{M}} \approx 2,600$), and $\mathcal{B}$ need only keep pace with K to hold queries-per-cell roughly constant, reaching $\approx 250\text{--}300\text{k}$ rather than the corpus size; $\mathcal{C}$ grows similarly modestly to $\approx 100\text{k}$. This asymmetry is deliberate: $\mathcal{A}$ requires only indexing, whereas each query in $\mathcal{B}$ and $\mathcal{C}$ costs one offline exact-KNN pass to label, so the expensive part of construction is bounded by the query sets ($\approx 350\text{--}400\text{k}$ passes at full scale) rather than by corpus size, and oversizing $\mathcal{B}$ beyond a well-populated cell floor spends exact-KNN effort for no accuracy gain.

Pilot reference space, and scale. At the pilot scale we are subsampling — choosing which 1M of the 7M pairs to include — so the stratification above is what keeps the subsample density-representative. For cross-model comparability the stratification and the resulting $\mathcal{A}/\mathcal{B}/\mathcal{C}$ assignment are computed *once*, in a single reference model’s embedding space, and the same pairs are reused for every model, so all models are evaluated on identical pairs in identical roles. The cost is that sparse-region coverage is guaranteed only in the reference space and is approximate in the others; we accept this at pilot scale, whose purpose is a go/no-go test, and disclose it rather than rely on it silently. At full corpus scale this consideration disappears: there is no subsampling, so no density-preserving stratification is required at all — $\mathcal{A}/\mathcal{B}/\mathcal{C}$ is a straight pair-level partition of the entire corpus, still shared across models, and any residual under-coverage of a particular model’s sparse cells is caught by the per-cell histogram check above, model by model, rather than by a cross-model stratification.

6.5 Why This Is Tractable Now

The one expensive step in Section 4.2 — the exact-KNN labeling pass — has already been run once per model as part of our prior large-scale evaluation [1], which measured $\text{CF}@100$ for exactly this index catalog on this corpus. Building and verifying the spatial map is therefore substantially a re-aggregation of already-collected labels by query location, rather than a new large-scale data-collection effort.

7 Building the Map: Offline Pipeline

We summarize the offline construction as a concrete pipeline over five stored artifacts, in the order they are produced.

1. **Landmarks.** Per model: landmark ID, centroid vector, and a corpus-only density score (for DO). $O(K)$ rows.
2. **Exact-KNN labels.** Per model: query ID, top-100 exact neighbor IDs. Computed once per model, $O(|Q|)$ rows, reused by every configuration built on that model.
3. **Query-landmark assignment.** Per model: query ID, nearest landmark ID, distance to nearest and second-nearest landmark. $O(|Q|)$ rows.
4. **Per-query fidelity.** Per configuration: query ID, $\text{CF}@k(q, A_j)$, obtained by running A_j against Q and overlapping with the exact-KNN labels. $O(|Q| \times m)$ rows — the largest table, but offline-only, never touched again once the next artifact exists.
5. **The served map.** Per configuration: landmark ID, mean, standard deviation, and count of $\text{CF}@k$ among queries assigned to that landmark. $O(K \times m)$ rows — small, and this is the only artifact (together with the landmark set itself) consulted at query time.

The gap between artifact 4 ($O(|Q| \times m)$, large, offline-only) and artifact 5 ($O(K \times m)$, small, served) is the entire practical point of the map: all per-query computation happens once, offline, and query-time serving touches only the small, pre-aggregated result.

8 Discussion: Intended Use

We scope this paper to building and verifying the map itself, not to the full system that would consume it. Concretely, once H1–H5 support H0, the map’s natural role is as a cheap, pre-execution signal for choosing an initial index configuration before any full search is run — deciding, per query, which configuration in a small catalog is predicted to meet a target fidelity level at the lowest cost. Designing and evaluating that downstream decision procedure, and combining the map with signals only available *after* a configuration has actually been queried, is deliberately left to separate work; this paper’s contribution is establishing whether the map’s core premise holds at all.

9 Limitations and Future Work

The verification methodology in Section 5 is designed to fail informatively: if H5 shows SM does not beat GA, or H1/H2 show error is unrelated to landmark distance, the right conclusion is that H0 does not hold at the tested scale or corpus, not that the map needs more tuning. We view this as a feature of the study design, not a limitation of it — but it does

mean the pipeline in Section 7 should be treated as producing a testable artifact, not a deployable one, until H1–H5 are run. We have also restricted this study to general-purpose corpora (LAION, MS-COCO, Flickr30K); whether H0 continues to hold under the sparser, more heterogeneous metadata and imagery of specialized or archival collections is an open question this study design does not address, and would need to be tested separately once such a corpus is available.

Finally, the decision-usefulness the map is ultimately meant to support (Section 1) is defined only relative to a fidelity floor τ . This study establishes whether the map predicts $CF@k$ accurately, and — via the swept-threshold metrics of Section 5 — whether that accuracy suffices to classify configurations against a range of candidate floors; it does not fix the floor that a deployed system would actually use. That floor follows from the full cost model — how latency, storage, and energy trade against fidelity in a given deployment — which is deliberately deferred to the parent work. Our threshold sweep is therefore a stand-in for a policy not yet specified: it shows across which floors the map is decision-useful, but confirming that the map suffices at the *particular* floor a real cost model implies is out of scope here.

10 Conclusion

Candidate fidelity is the standard way of quantifying what an approximate index gives up relative to exact search, but every existing way of measuring it requires the exact computation approximate indexing exists to avoid. We proposed the spatial fidelity map as a query-time, ground-truth-free alternative built once, offline, and consulted via a single nearest-landmark lookup, and we specified a complete, self-contained methodology — five targeted experiments and a concrete storage-level construction pipeline — for verifying whether the map’s underlying premise actually holds before it is trusted in a live system.

References

- [1] Anonymous Authors. Scalability analysis of cross-modal text-to-image pipelines. Under submission, 2026.
- [2] Wei Chen, Jincai Chen, Fuhao Zou, Yuan-Fang Li, Ping Lu, Qiang Wang, and Wei Zhao. Vector and line quantization for billion-scale similarity search on gpus. *Future Gener. Comput. Syst.*, 99:295–307, 2019.
- [3] Tiezheng Ge, Kaiming He, Qifa Ke, and Jian Sun. Optimized product quantization. *IEEE Trans. Pattern Anal. Mach. Intell.*, 36(4):744–755, 2014.
- [4] Hervé Jégou, Matthijs Douze, and Cordelia Schmid. Product quantization for nearest neighbor search. *IEEE Trans. Pattern Anal. Mach. Intell.*, 33(1):117–128, 2011.
- [5] Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with gpus. *CoRR*, abs/1702.08734, 2017.
- [6] Ravishankar Krishnaswamy, Magdalen Dobson Manohar, and Harsha Vardhan Simhadri. The diskann library: Graph-based indices for fast, fresh and filtered vector search. *IEEE Data Eng. Bull.*, 48(3):20–42, 2024.
- [7] Tsung-Yi Lin, Michael Maire, Serge J. Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft COCO: common objects in context. In David J. Fleet, Tomás Pajdla, Bernt Schiele, and Tinne Tuytelaars, editors, *Computer Vision - ECCV 2014 - 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V*, volume 8693 of *Lecture Notes in Computer Science*, pages 740–755. Springer, 2014.
- [8] Yury A. Malkov and Dmitry A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(4):824–836, 2020.
- [9] James Jie Pan, Jianguo Wang, and Guoliang Li. Survey of vector database management systems. *VLDB J.*, 33(5):1591–1615, 2024.
- [10] Christoph Schuhmann, Richard Vencu, Romain Beaumont, Robert Kaczmarczyk, Clayton Mullis, Aarush Katta, Theo Coombes, Jenia Jitsev, and Aran Komatsuzaki. LAION-400M: open dataset of clip-filtered 400 million image-text pairs. *CoRR*, abs/2111.02114, 2021.
- [11] Peter Young, Alice Lai, Micah Hodosh, and Julia Hockenmaier. From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. *Trans. Assoc. Comput. Linguistics*, 2:67–78, 2014.